



GIRRAWEE HIGH SCHOOL

YEAR 11 - TASK 2

2006

MATHEMATICS

2 UNIT

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new sheet of paper.

Question 1 (13 marks)

- (a) Find $\sqrt[4]{1.6 \times 2.6}$ correct to 3 decimal places. 1
- (b) Evaluate $\frac{13^5}{11^6 + 17^4}$ correct to 3 significant figures. 1
- (c) Write 0.01072 in scientific notation. 1
- (d) Find the value of x if $\sqrt{18} + \sqrt{8} = \sqrt{x}$ 2
- (e) Classify each of these real numbers as rational.
 $\sqrt{9}, \sqrt{10}, \sqrt[3]{15}, -0.16, \pi$ 2
- (f) Write each recurring decimal as a fraction in lowest terms.
- (i) $0.3\dot{6}$ 3
- (ii) $0.\dot{2}9\dot{7}$ 3

Question 2 (19 marks)

(a) Simplify.

(i) $\sqrt{500}$

2

(ii) $3\sqrt{2} + 3\sqrt{8} - \sqrt{50}$

2

(b) Expand and simplify.

(i) $(2\sqrt{3} - 1)(3\sqrt{3} + 5)$

2

(ii) $(5\sqrt{2} - 3)^2$

3

(c) Express with a rational denominator.

(i) $\frac{5}{2\sqrt{7}}$

2

(ii) $\frac{3}{\sqrt{11} + \sqrt{5}}$

3

(d) Solve for x .

(i) $|2x - 5| = 3$

2

(ii) $|2 + 4x| \geq 6$

3

Question 3 (19 marks)

- (a) Expand and simplify
- (i) $-4x^2(x+3) - 2x^2(x-1)$ 2
- (ii) $(3x-5)(2x-3)$ 2
- (b) Factorise
- (i) $36 - 25g^2$ 2
- (ii) $i^2 + 5i - 36$ 2
- (iii) $3k^2 - 7k - 6$ 3
- (iv) $n^3 + 8$ 3
- (v) $u^2w + vw - u^2x - vx$ 3
- (c) Find the values of p and q such that $\frac{\sqrt{5}}{\sqrt{5}-2} = p + q\sqrt{5}$. 2

Question 4 (15 marks)

(a) Simplify

(i) $\frac{3a}{2b} + \frac{2a}{3b}$

2

(ii) $\frac{2}{x+1} - \frac{5}{x-4}$

3

(b) Factor where possible, then simplify

(i) $\frac{2x-2y}{x^2-y^2}$

2

(ii) $\frac{3x^2-19x-14}{9x^2-4}$

3

(c) Solve each linear equation

(i) $3(x+5) = 17$

2

(ii) $\frac{a}{7} - 4 = \frac{a}{2} + 11$

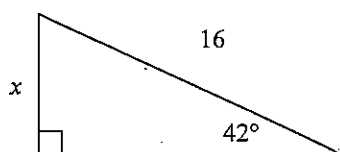
3

Question 5 (14 marks)

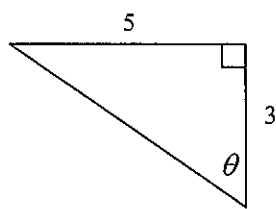
(a) Find correct to 4 decimal places, $\sin 38^\circ 24'$. 1

(b) Find the acute angle θ , correct to the nearest minute,
given that, $\cos \theta = \frac{1}{4}$. 2

(c) Find, correct to two decimal places, the side marked x . 2

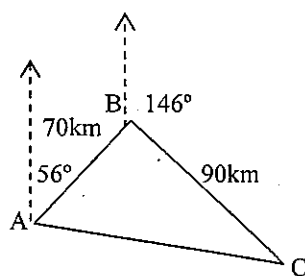


(d) Find, correct to the nearest minute, the angle θ . 2



(e) A vertical pole stands on level ground. From a point on the ground 8 metres from its base, the angle of elevation of the top of the pole is 38° . Find the height of the pole, correct to the nearest centimeter. 2

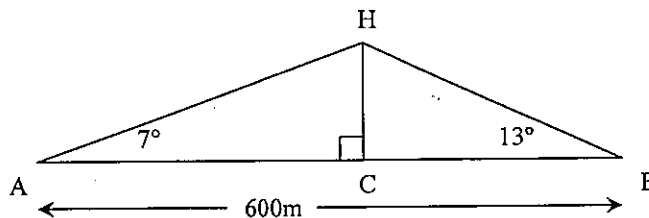
(f) A motorist drove 70 km from town A to town B on a bearing of 056° , and then drove 90km from town B to town C on a bearing of 146° .



- (i) Explain why $\angle ABC = 90^\circ$ 1
- (ii) How far apart are the towns A and C, correct to the nearest metre 2
- (iii) Find $\angle BAC$, and hence find the bearing of town C from A, correct to the nearest degree. 2

Question 6 (23 marks)

- (a) Find the exact value of
- | | |
|-----------------------|---|
| (i) $\tan 240^\circ$ | 1 |
| (ii) $\sin 315^\circ$ | 1 |
- (b) Simplify $\frac{1}{\cot \theta}$ 1
- (c) Prove the following trigonometric identities
- | | |
|---|---|
| (i) $\tan \theta \operatorname{cosec} \theta = \sec \theta$ | 2 |
| (ii) $4 \sec^2 \theta - 3 = 1 + 4 \tan^2 \theta$ | 2 |
- (d) Solve for $0^\circ \leq x \leq 360^\circ$.
- | | |
|-------------------------------|---|
| (i) $\sqrt{2} \sin x + 1 = 0$ | 2 |
| (ii) $\cos 2x = \frac{1}{2}$ | 3 |
| (iii) $\sin^2 x + \sin x = 0$ | 3 |
- (e) A triangle has sides 7cm, 8cm and 10cm. Use the cosine rule to find its largest angle, and hence find the area of the triangle, correct to the nearest cm^2 . 4
- (f) A helicopter is hovering above a straight, horizontal road AB of length 600 metres. The angles of elevation of H from A and B are 7° and 13° respectively. The point C lies on the road directly below H.



- | | |
|---|---|
| (i) Use the sine rule to show that $HB = \frac{600 \sin 7^\circ}{\sin 160^\circ}$ | 2 |
| (ii) Hence find the height CH of the helicopter above the road, correct to the nearest metre. | 2 |

Yr 11 - Mathematics - Task 2 (2006)

Q1 a) $\sqrt[4]{1.6 \times 2.6} = 1.428$ ①

b) $\frac{13^5}{11^6 + 17^4} = 0.200$ ①

c) $0.01072 = 1.072 \times 10^{-2}$ ①

d) $\sqrt{18} + \sqrt{8} = 3\sqrt{2} + 2\sqrt{2} = 5\sqrt{2}$ ②

e) Rational no's. are $\sqrt{9}, -0.16$ ②

Ans $x = 0.3\dot{6}$

$10x = 3.666...$

$100x = 36.666...$

$100x - 10x = 36.66... - 3.66...$

$90x = 33$

$x = \frac{33}{90} = \frac{11}{30}$ ③

ii) $x = 0.29\dot{7}$

$x = 0.297297...$

$1000x = 297.297297...$

$1000x - x = 297.297... - 0.297...$

$999x = 297$

$x = \frac{297}{999} = \frac{11}{37}$ ③

Q2 a) i) $\sqrt{500} = \sqrt{100 \times 5} = 10\sqrt{5}$ ②

ii) $3\sqrt{2} + 3\sqrt{8} - \sqrt{50}$
 $= 3\sqrt{2} + 6\sqrt{2} - 5\sqrt{2}$
 $= 4\sqrt{2}$ ②

b) i) $(2\sqrt{3} - 1)(3\sqrt{3} + 5)$
 $= 6\sqrt{9} + 10\sqrt{3} - 3\sqrt{3} - 5$
 $= 18 + 7\sqrt{3} - 5$
 $= 13 + 7\sqrt{3}$ ②

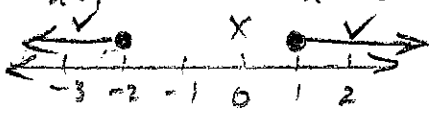
ii) $(5\sqrt{2} - 3)^2$
 $= 25\sqrt{4} - 30\sqrt{2} + 9$
 $= 59 - 30\sqrt{2}$ ③

c) i) $\frac{5}{2\sqrt{7}} = \frac{5}{2\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$
 $= \frac{5\sqrt{7}}{14}$ ②

ii) $\frac{3}{\sqrt{11} + \sqrt{5}} \times \frac{\sqrt{11} - \sqrt{5}}{\sqrt{11} - \sqrt{5}}$
 $= \frac{3\sqrt{11} - 3\sqrt{5}}{11 - 5}$
 $= \frac{\sqrt{11} - \sqrt{5}}{2}$ ③

d) i) $2x - 5 = 3$ $2x - 5 = -3$
 $2x = 8$ $2x = 2$
 $x = 4$ $x = 1$ ②

ii) $2 + 4x = 6$ $2 + 4x = -6$
 $4x = 4$ $4x = -8$
 $x = 1$ $x = -2$



$\therefore x \leq -2, x \geq 1$ ③

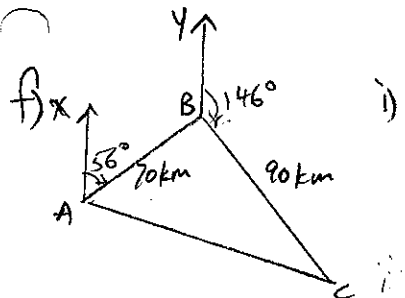
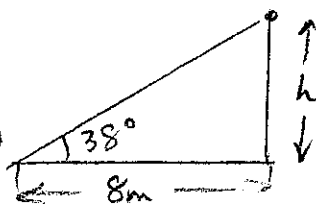
Q5 a) $\sin 38^\circ 24' = 0.6211$. (1)

b) $\cos \theta = \frac{1}{4}$
 $\theta = \cos^{-1}(\frac{1}{4})$
 $\theta = 75^\circ 31'$ (2)

c) $\sin 42^\circ = \frac{x}{16}$
 $x = 16 \sin 42^\circ$
 $x = 10.7060897\dots$ (2)
 $x = 10.71$ (to 2 dec pl.)

d) $\tan \theta = \frac{5}{3}$
 $\theta = \tan^{-1}(\frac{5}{3})$
 $\theta = 59^\circ 02'$ (2)

e) $\tan 38^\circ = \frac{h}{8}$
 $h = 8 \tan 38^\circ$
 $h = 6.25 \text{ m}$ (2)



i) $\angle ABy = 124^\circ$ (co-int. $\angle$'s. // lines)
 $\angle ABC = 360 - 124 - 146$ (revolution)
 $\angle ABC = 90^\circ$ (1)

ii) Since $\angle ABC = 90^\circ$
 $70^2 + 90^2 = AC^2$
 $AC^2 = 13000$
 $AC = \sqrt{13000}$ (2)
 $AC = 114.018 \text{ m}$ to nearest
 $\approx 114.02 \text{ km}$ (metre)

iii) $\tan \angle BAC = \frac{90}{70}$
 $\angle BAC = 52^\circ$ (to nearest degree)

$\therefore$ Bearing A to C
 $= 56 + 52$
 $= 108^\circ$ (2)

Q6 a) i) $\tan 240^\circ = \tan 60^\circ$
 $= \sqrt{3}$. (1)

ii) $\sin 315^\circ = -\sin 45^\circ$
 $= -\frac{1}{\sqrt{2}}$ (1)

b) $\frac{1}{\cot \theta} = \frac{1}{\frac{1}{\tan \theta}} = \tan \theta$ (1)

c) i) $\tan \theta \operatorname{cosec} \theta = \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\sin \theta}$
 $= \frac{1}{\cos \theta}$
 $= \sec \theta$. (2)

ii) $4 \sec^2 \theta - 3 = 4(\tan^2 \theta + 1) - 3$
 $= 4 \tan^2 \theta + 4 - 3$
 $= 1 + 4 \tan^2 \theta$ (2)

d) i) $\sqrt{2} \sin x + 1 = 0$
 $\sin x = -\frac{1}{\sqrt{2}}$

$x = 225^\circ, 315^\circ$ (2)

ii) $\cos 2x = \frac{1}{2}$ $\begin{cases} 0 \leq x \leq 360 \\ 0 \leq 2x \leq 720 \end{cases}$

$\therefore 2x = 60^\circ, 300^\circ, 420^\circ, 660^\circ$
 $x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

(3)

iii) $\sin^2 x + \sin x = 0$
 $\sin x (\sin x + 1) = 0$

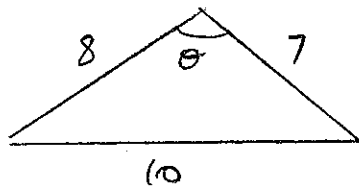
$\therefore \sin x = 0$ or $\sin x = -1$

$\therefore x = 0, 180^\circ, 360^\circ$ or $x = 270^\circ$

$\therefore x = 0, 180^\circ, 270^\circ, 360^\circ$

(3)

(Q6 e)



$$\cos \theta = \frac{8^2 + 7^2 - 10^2}{2(8)(7)}$$

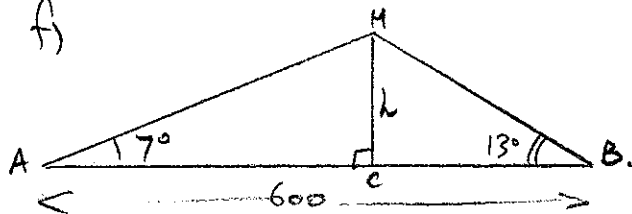
$$\cos \theta = \frac{13}{112}$$

$$\theta = 83^\circ 20' 04''$$

$$\therefore \text{Area} = \frac{1}{2}(8)(7)\sin 83^\circ 20' 04''$$
$$= 27.81 \text{ cm}^2$$

$$\text{Area} \approx 28 \text{ cm}^2 \text{ (to nearest cm}^2\text{)} \quad (4)$$

f)



$$\therefore \angle AHB = 180 - 13 - 7$$
$$= 160^\circ$$

$$\text{i)} \quad \frac{HB}{\sin 7^\circ} = \frac{600}{\sin 160^\circ}$$

$$\therefore HB = \frac{600 \sin 7^\circ}{\sin 160^\circ} \quad (2)$$

$$\text{ii)} \quad \therefore \sin 13^\circ = \frac{h}{HB}$$

$$h = HB \sin 13^\circ$$

$$h = \frac{600 \sin 7^\circ \sin 13^\circ}{\sin 160^\circ}$$

$$h = 48.09 \dots$$

$$h \approx 48 \text{ m (to nearest metre)} \quad (2)$$

$$\textcircled{Q3} \text{ a) i) } -4x^2(x+3) - 2x^2(x-1) \\ = -4x^3 - 12x^2 - 2x^3 + 2x^2 \\ = -6x^3 - 10x^2 \quad \textcircled{2}$$

$$\text{ii) } (3x-5)(2x-3) \\ = 6x^2 - 9x - 10x + 15 \\ = 6x^2 - 19x + 15 \quad \textcircled{2}$$

$$\text{b) i) } 36 - 25g^2 = (6-5g)(6+5g) \quad \textcircled{2}$$

$$\text{ii) } i^2 + 5i - 36 = (i+9)(i-4) \quad \textcircled{2}$$

$$\text{iii) } 3k^2 - 7k - 6 = (3k+2)(k-3) \quad \textcircled{3}$$

$$\begin{array}{r|l} 3k & +2 \\ \times & \\ k & -3 \\ \hline & -9k + \\ & -7k \end{array}$$

$$\text{iv) } n^3 + 8 = (n+2)(n^2 - 2n + 4) \quad \textcircled{3}$$

$$\text{v) } u^2w + vw - u^2x - vx \\ = w(u^2 + v) - x(u^2 + v) \\ = (u^2 + v)(w - x) \quad \textcircled{3}$$

$$\text{c) } \frac{\sqrt{5}}{\sqrt{5}-2} \times \frac{(\sqrt{5}+2)}{(\sqrt{5}+2)} = \frac{5+2\sqrt{5}}{5-4} \\ = \frac{5}{\sqrt{5}-2} = 5+2\sqrt{5}$$

$$\therefore p=5, q=2 \quad \textcircled{2}$$

$$\textcircled{Q4} \text{ a) i) } \frac{3a}{2b} + \frac{2a}{3b} = \frac{9a}{6b} + \frac{4a}{6b} \\ = \frac{13a}{6b} \quad \textcircled{2}$$

$$\text{ii) } \frac{2}{x+1} - \frac{5}{x-4} = \frac{2(x-4) - 5(x+1)}{(x+1)(x-4)} \\ = \frac{2x-8-5x-5}{(x+1)(x-4)} \\ = \frac{-3x-13}{(x+1)(x-4)} \quad \textcircled{3}$$

$$\text{b) i) } \frac{2x-2y}{x^2-y^2} = \frac{2(x-y)}{(x+y)(x-y)} \\ = \frac{2}{x+y} \quad \textcircled{2}$$

$$\text{ii) } \frac{3x^2 - 19x - 14}{9x^2 - 4} = \frac{(3x+2)(x-7)}{(3x+2)(3x-2)} \\ = \frac{x-7}{3x-2} \quad \textcircled{3}$$

$$\text{c) i) } 3(x+5) = 17 \\ 3x + 15 = 17 \\ 3x = 2 \\ x = \frac{2}{3} \quad \textcircled{2}$$

$$\text{ii) } \frac{a}{7} - 4 = \frac{a}{2} + 11 \quad (\times 14)$$

$$2a - 56 = 7a + 154 \\ 5a = -210 \\ a = -42 \quad \textcircled{3}$$